



DATA ANALYTICS PROJECT · POWER BI

IT Support Dashboard

Analysis of 1,809 help desk tickets: from data cleaning to an interactive dashboard with SLA and satisfaction KPIs.

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Context and goal

An IT help desk receives tickets from 8 business areas, but there was no visibility into volumes, SLA compliance, or user satisfaction.



Goal

Build an executive Power BI dashboard that centralizes the state of the support operation.



Business questions

Are we meeting SLAs? Which categories and areas generate the most tickets? How does volume evolve month over month?



Tools

Excel as the source, Power Query for cleaning, DAX for measures, and Power BI Desktop for visualization.

The dataset

An Excel file with 5 related tables — a typical help desk data model.

1,809 tickets logged (year 2025)

17 columns per ticket

12 agents across 3 offices

8 requesting business areas



Tables in the model

Tickets — facts: dates, SLA, satisfaction

Agents — team, seniority, and office

Areas — catalog of requesting areas

SLA_Priorities — target hours by priority

Categories — incident classification

Cleaning and shaping in Power Query

The raw data came with real quality issues that had to be solved before modeling:



Inconsistent text casing

“conectividad”, “CONECTIVIDAD” and “Conectividad” appeared as different categories. Fixed with capitalization and Trim.



Duplicate concept values

“Acceso y permisos” vs “Accesos y permisos”, “RRHH” vs “Recursos Humanos”, “Teams” vs “Microsoft Teams”. Unified with value replacement.



Meaningful nulls

Open tickets had no close date or resolution hours. They were kept and handled inside the DAX measures.



Data types

Dates and times stored as text; typed correctly to enable time intelligence and the monthly axis.

Data model and DAX measures



Star schema

The Tickets fact table relates to the Agents, Areas, Categories and SLA_Priorities dimensions through 1:N relationships, enabling the priority and agent slicers in the report.



Calculated KPIs

Measures using CALCULATE, DIVIDE and AVERAGE to count tickets by status, compute the SLA compliance rate, and average user ratings.

1,809

Total Tickets

342

Open tickets

1,467

Closed tickets

48.40 %

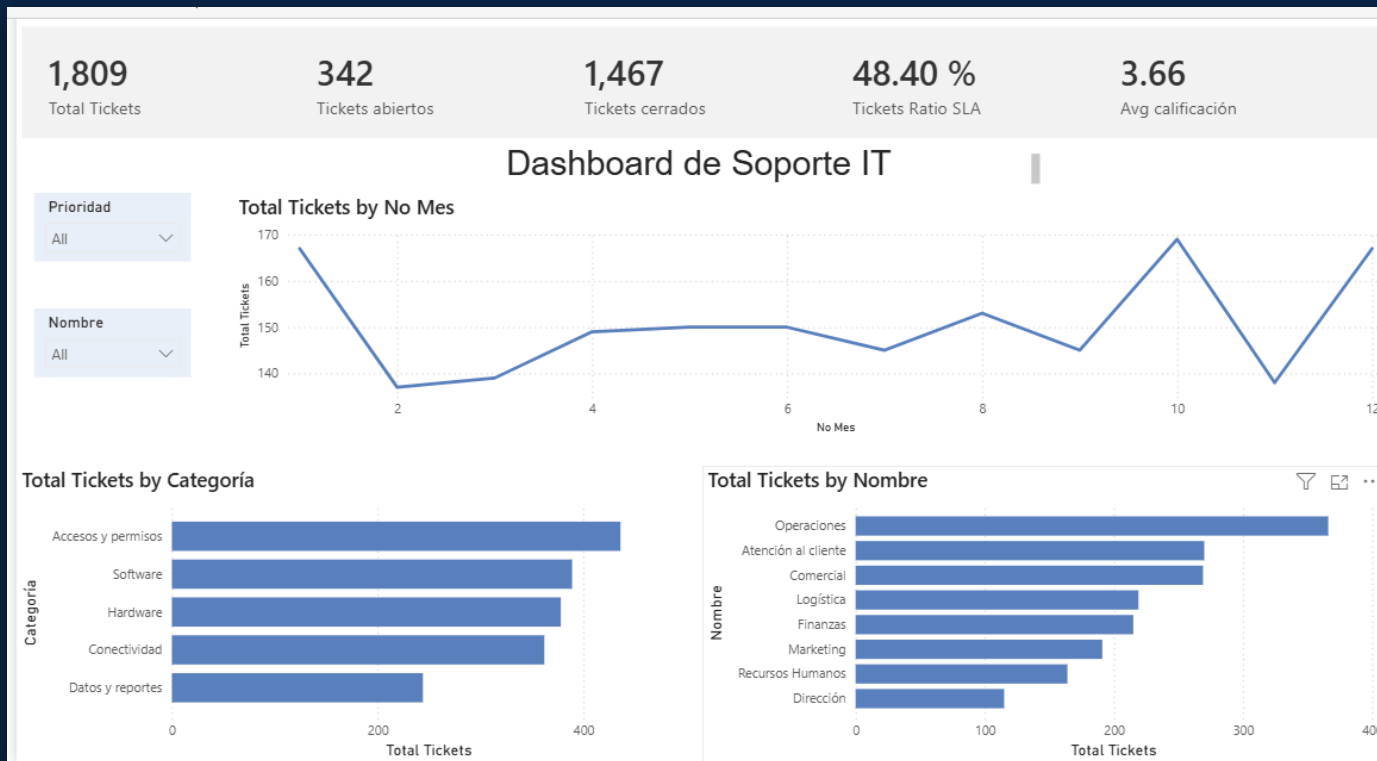
SLA compliance

3.66 / 5

Average rating

The dashboard

A single interactive page with slicers by priority and by agent.



Key insights



SLA compliance is the critical issue

Only 48.4% of closed tickets met their target time. Average resolution (62 h) often exceeds the 24–48 h targets for high and medium priorities.



Operations drives the demand

With 366 tickets it is the top requesting area, followed by Customer Service and Sales (269 each). Management has the lowest volume.



Four categories dominate

Software, Hardware, Connectivity, and Access & permissions generate ~80% of the volume; “Data & reports” is the smallest but has automation potential.



Satisfaction has room to improve

Average rating of 3.66/5. Monthly volume is stable (137–169 tickets), with peaks in January, October, and December.

Conclusions and next steps

The project covers the full analytics cycle: raw data → cleaning → modeling → KPIs → decisions.



Business recommendation

Prioritize a review of the resolution process: reinforce Level 1 support in high-volume categories to lift SLA compliance above 70%.



Report improvements

Add analysis by agent and office, a monthly SLA trend, and a tooltip comparing resolution hours vs. target.



Next iterations

Connect the source to a SQL database, schedule automatic refresh, and publish to the Power BI service.